PAPER ID-310610

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VII) THEORY EXAMINATION 2025-26
MEASURMENT & METROLOGY

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 07 = 14**

Q no.	Question	CO
a.	What is meant by sensitivity and stability of a measuring instrument?	1
b.	Explain static and dynamic response of a measuring instrument.	1
c.	What are the common methods to measure linear displacement?	1
d.	Define a strain rosette and its purpose.	2
e.	List any three linear measurement devices used in workshops.	2
f.	What are limit gauges? How are they used in manufacturing?	1
g.	Define surface roughness. How is it quantitatively evaluated?	2

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Explain the concept of generalized measurement system and discuss the role of each element.	2
b.	Explain the working principle of a strain gauge. How does it convert mechanical strain into electrical signal?	2
c.	Describe the working of Laser Doppler Velocimetry (LDV) and explain how it differs from traditional flow measurement techniques.	3
d.	Describe the concept of interchangeability and standardization. How do they affect manufacturing quality?	3
e.	Describe methods for measurement of screw threads and gears, highlighting key instruments used.	4

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Discuss types of transducers and their characteristics with suitable examples.	2
b.	Explain the importance of statistical analysis of error in evaluating measurement systems.	2

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Describe the method of frequency measurement by direct comparison. What are its advantages and limitations?	2
b.	List and explain the different methods of displacement measurement. Which method is most suitable for micro-displacement?	2

5. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Discuss the construction and working principle of a bimetallic thermometer. What are its common industrial applications?	2
b.	Explain the use of pneumatic and hydraulic systems for measuring force and torque. What are their advantages and limitations?	2

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6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Discuss the Johansson's Microkrator and its role in precision measurement.	3
b.	Describe various linear and angular measurement systems, highlighting their accuracy and typical applications.	3

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Discuss the quantitative evaluation of surface texture and the different parameters used for measurement.	4
b.	Explain how surface roughness affects the performance of mechanical components, giving two practical examples.	4

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